

Introduction and Taxonomy of Uncertainty Quantification

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Outline

- Why estimate uncertainty?
- Sources of uncertainty
- Approaches to uncertainty estimation
- Evaluation metrics & calibration



Cambridge
Dictionary

“a situation in which something is not known”

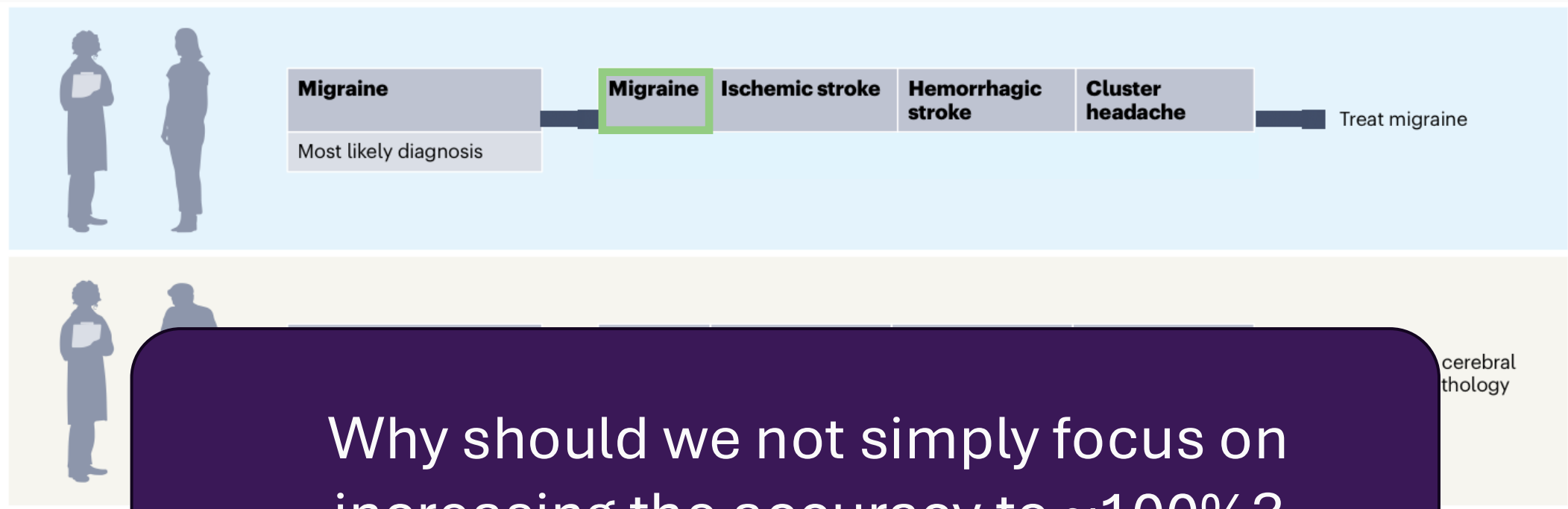


“situations involving imperfect or unknown information”

Uncertainty refers to the measure of confidence or lack of certainty in a prediction.

Why estimate uncertainty?

Scenarios:



Why should we not simply focus on increasing the accuracy to ~100%?

Why estimate uncertainty?

- **Failure detection:** when not to trust the prediction
- **Interpretability:** e.g. uncertainty maps
- **Data efficiency:** e.g. active learning
- **Monitoring:** post-deployment QC
- **Regulation:** fairness audits, compliance

→ Safer, more trustworthy models

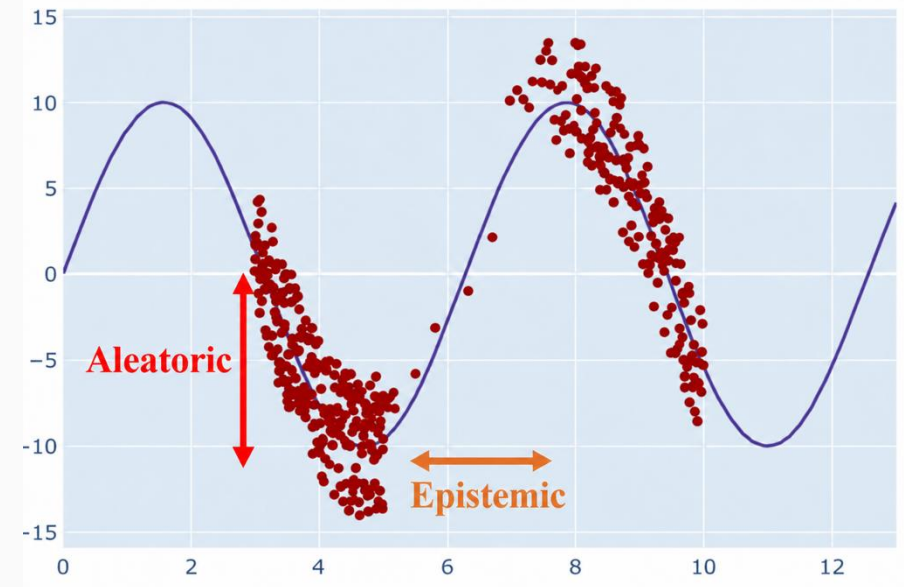
Sources of uncertainty

Aleatoric uncertainty:

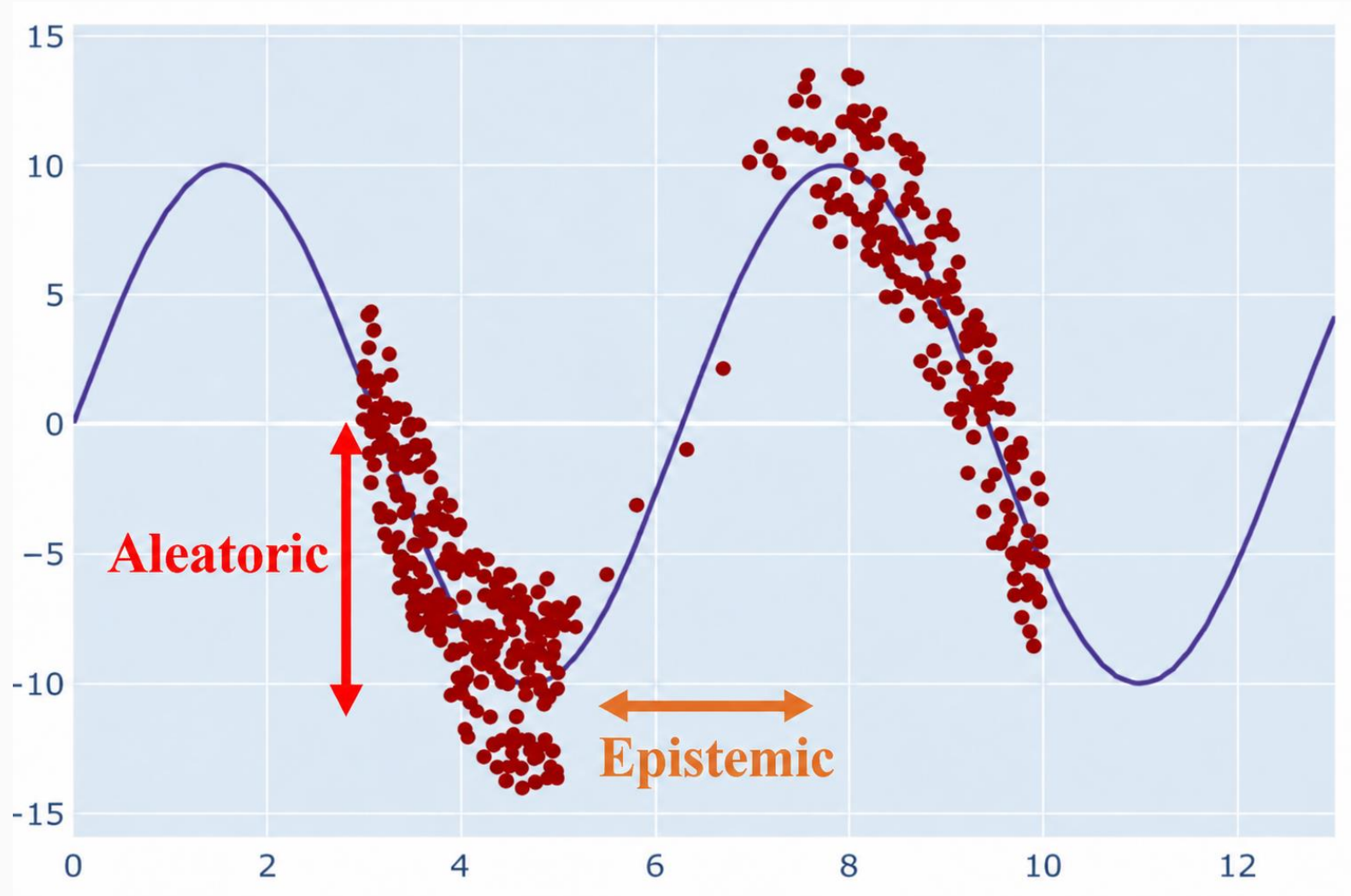
→ ‘Data-inherent’ uncertainty

Epistemic uncertainty:

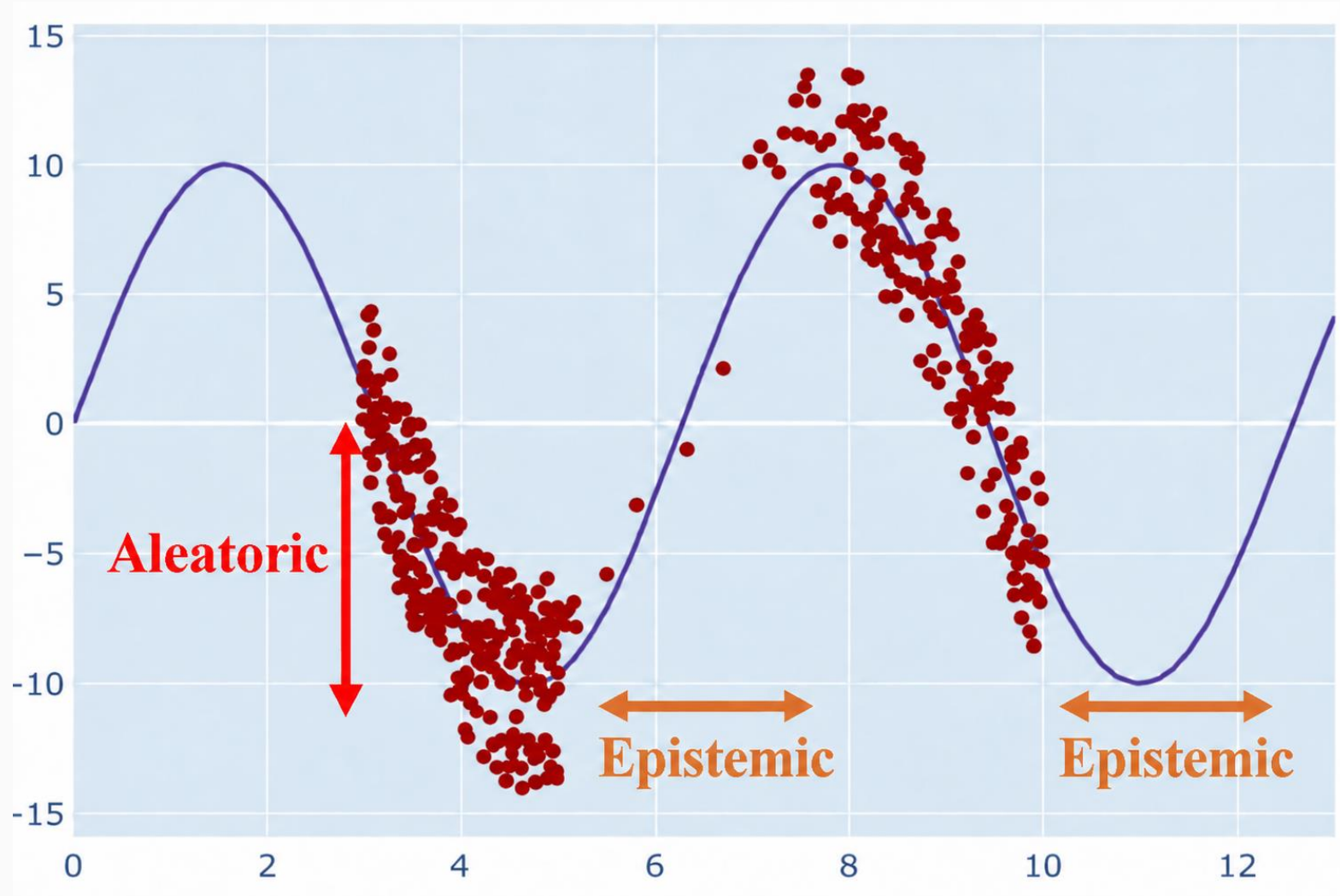
→ ‘Lack of knowledge’ uncertainty



Sources of uncertainty



Sources of uncertainty



Aleatoric or epistemic?



Aleatoric

Aleatoric or epistemic?



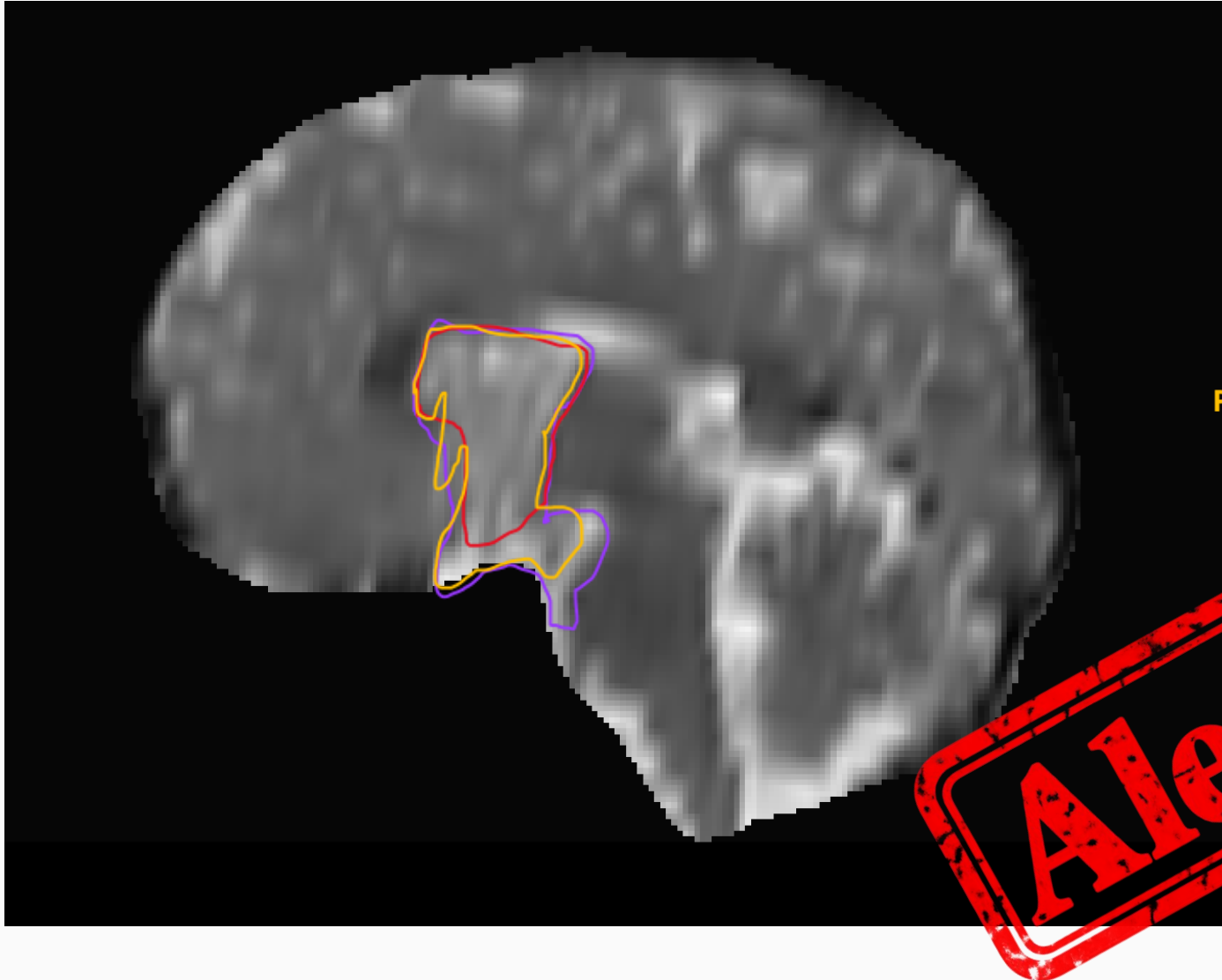
By Paola Nogales Ascarrunz, CC BY-SA 4.0, <https://commons.wikimedia.org/w/index.php?curid=199817826>

Aleatoric or epistemic?

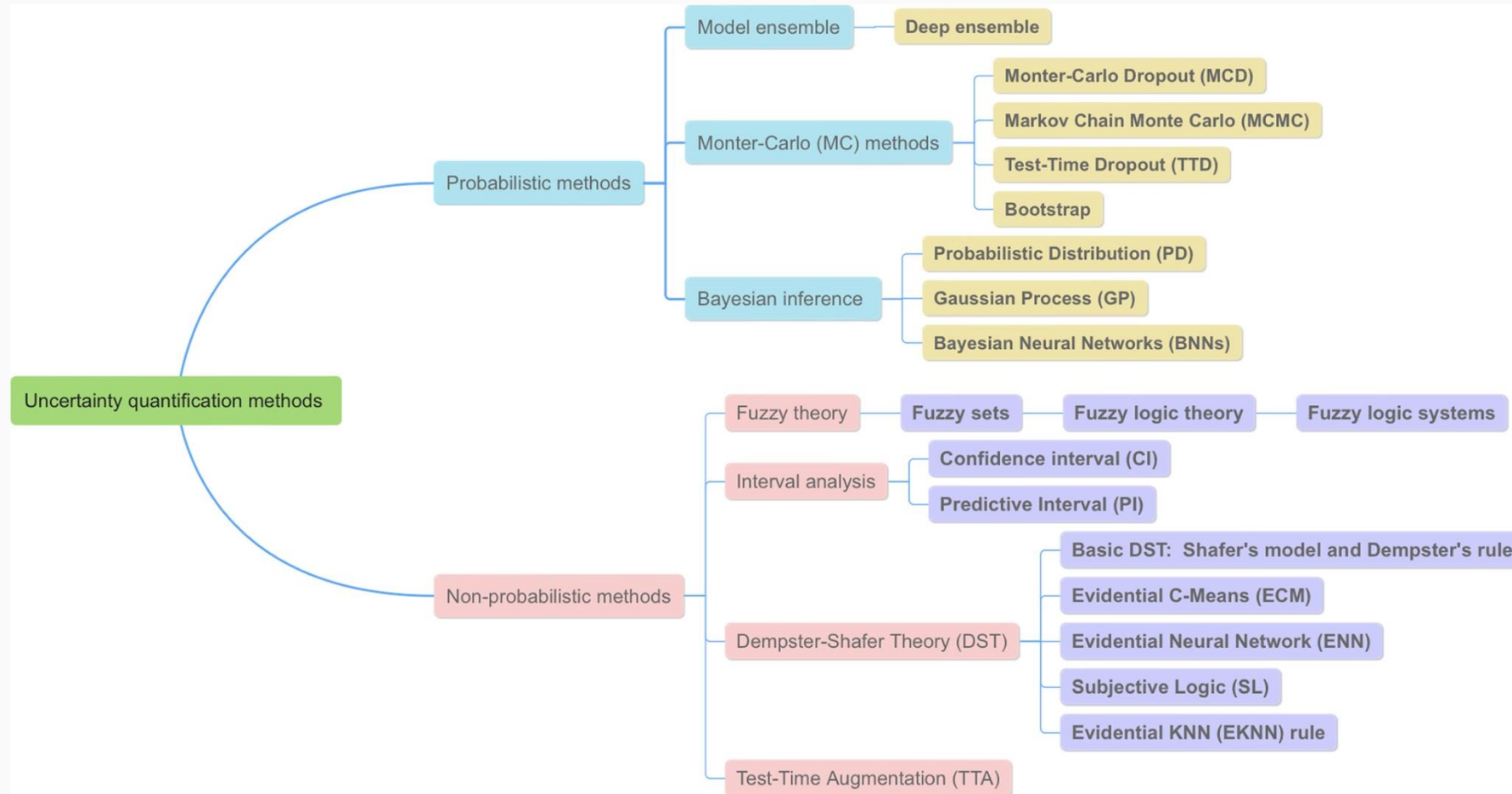
Tracer #	Scheltens 0		Scheltens 2		Scheltens 4	
	Local	Harmonized	Local	Harmonized	Local	Harmonized
6						
10						
13						

Epistemic

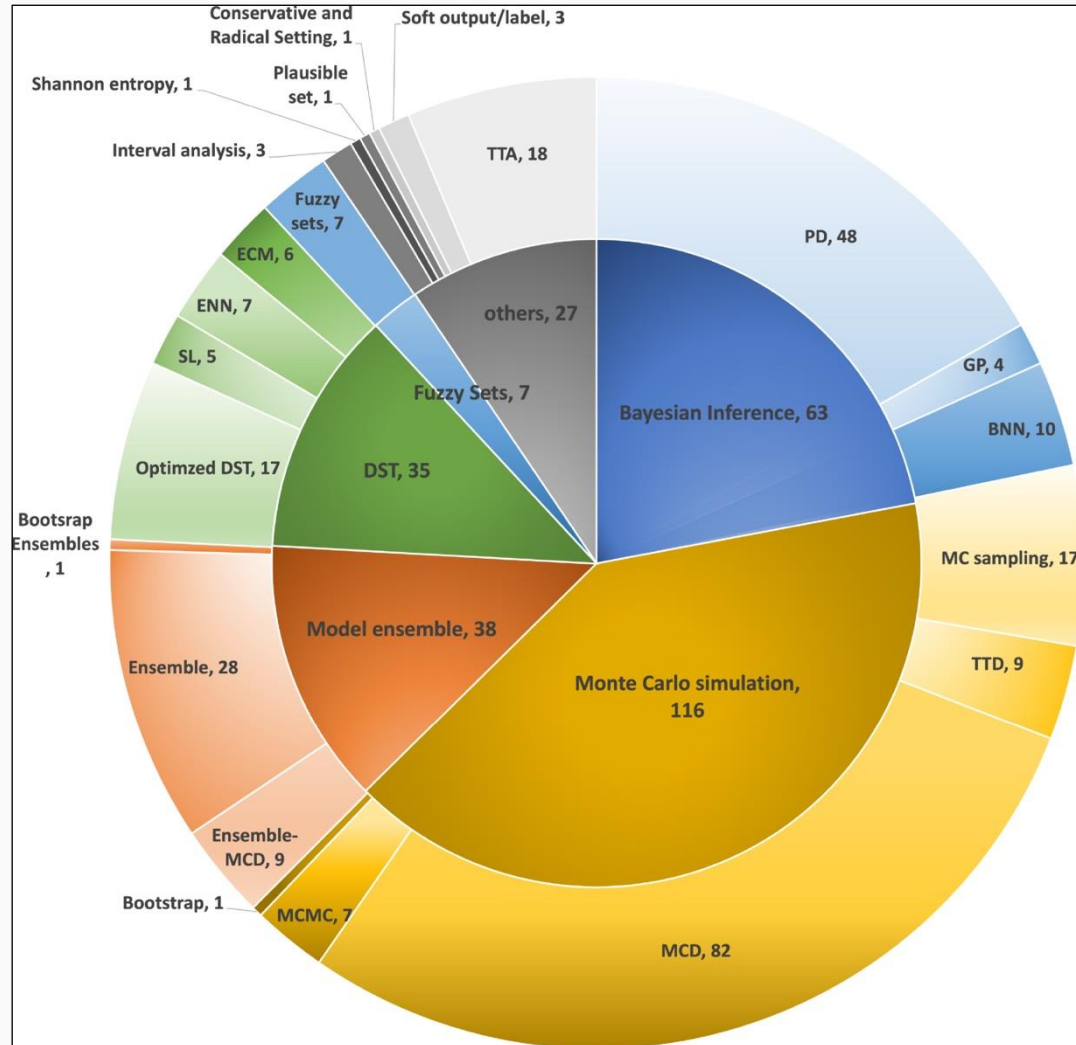
Aleatoric or epistemic?



Approaches to uncertainty estimation



Approaches to uncertainty estimation



Evaluation metrics & calibration

1

Proper scoring rules

Combining predictive accuracy *and* calibration

Examples: Negative Log-Likelihood (NLL), Brier Score

$$\text{NLL} = -\frac{1}{N} \sum_{i=1}^N \log p_{i,y_i}$$

$$\text{BS} = \frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C (p_{i,c} - 1[y_i = c])^2$$

N : Number of samples (e.g., voxels)

C : Number of classes

p_{i,y_i} : Predicted probability assigned to the true class y_i

$p_{i,c}$: Predicted probability of class c for sample i

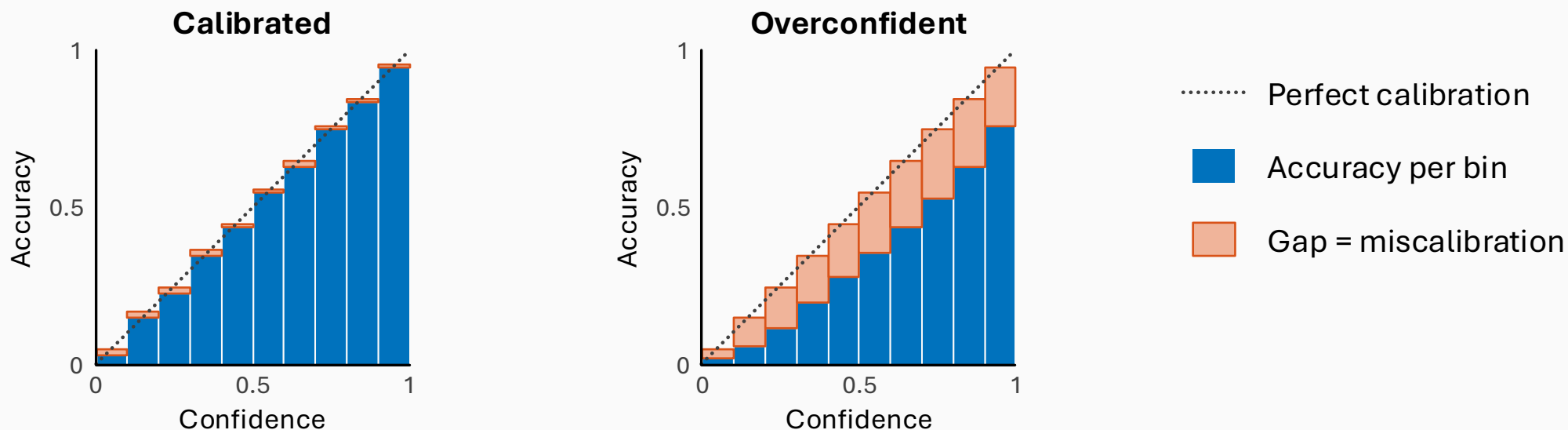
y_i : Ground-truth class for sample i

Evaluation metrics & calibration

2

Calibration

“Does confidence reflect actual error rate?”



$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{n} |\text{acc}(B_m) - \text{conf}(B_m)|$$

$$\text{AECE} = \frac{1}{M} \sum_{m=1}^M |\text{acc}(B_m) - \text{conf}(B_m)|$$

Evaluation metrics & calibration

3

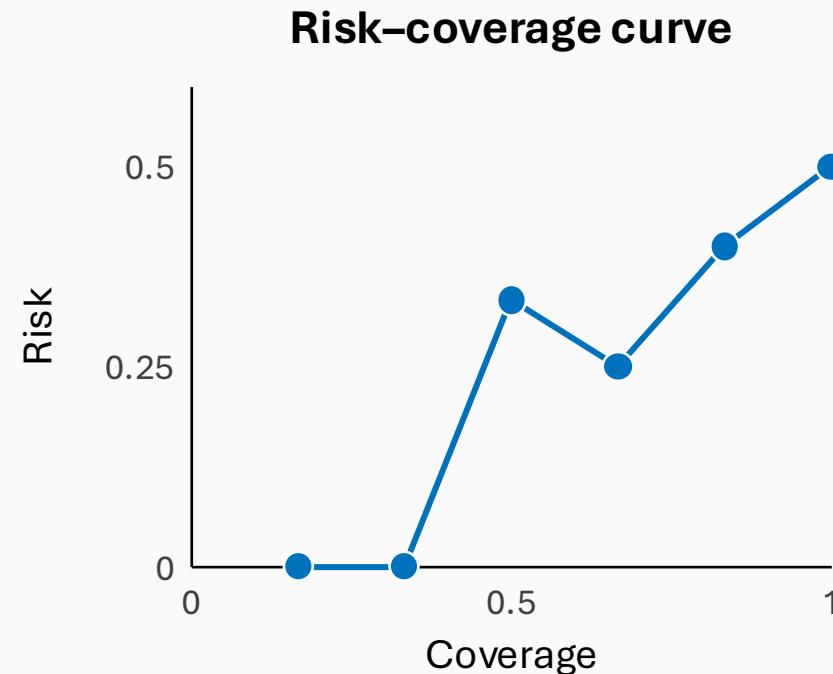
Downstream utility

Assessing usefulness

Examples: Risk-coverage curve

Pred.	Confidence	Correct?	Risk
A	0.62	✗	0/1 = 0.00
B	0.95	✓	0/2 = 0.00
C	0.40	✗	1/3 = 0.33
D	0.88	✓	1/4 = 0.25
E	0.71	✓	2/5 = 0.40
F	0.83	✗	3/6 = 0.50

risk = false predictions / all predictions



Summary

- **Uncertainty captures how confident model is**
- **Can increase safety and trustworthiness**
- **Some uncertainties are irreducible**
- **Evaluation: scoring rules, calibration, risk-coverage**

Questions?

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